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Project Title

MindGuard:

Emotion-Aware and Crisis-Sensitive NLP Support System

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1. Executive Summary

MindGuard is an advanced artificial intelligence prototype that demonstrates the integration of multiple cutting-edge AI techniques for mental health support applications. The system addresses a critical real-world need: the gap between mental health support demand and the availability of trained professionals, particularly in regions such as Sri Lanka where the ratio of mental health professionals to population remains critically low(World Health Organization, 2022).

The system integrates three core AI components:

- (1) multi-label emotion classification using a fine-tuned DistilBERT transformer model (Sanh et al., 2019) trained on the GoEmotions dataset (Demszky et al., 2020) across 28 emotional labels;
- (2) crisis-risk detection using a Variational Autoencoder (Kingma & Welling, 2013) trained as an unsupervised anomaly detector on non-crisis text;
- (3) empathetic response generation leveraging systematic prompt engineering strategies -zero-shot, few-shot, and chain-of-thought (Wei et al., 2022) ; with the Llama 3.1 8B model via the Groq API.

The emotion classification model achieved a test micro-F1 score of 0.575 and macro-F1 of 0.418 across 28 emotion labels after three training epochs on 43,410 GoEmotions samples. The VAE-based crisis detector successfully flags explicit crisis language through a hybrid approach combining reconstruction error anomaly detection with keyword safety enforcement. The system is deployed as a FastAPI backend with a human-centred Gradio web interface featuring a breathing exercise, rotating affirmations, and calming nature imagery.

The project fulfils all five required technique categories from the Advanced AI module specification: NLP preprocessing, transformer-based transfer learning, generative AI (VAE), prompt engineering, and transfer learning. All training runs are tracked with MLflow for full reproducibility.

2. Problem Statement and Motivation

2.1 Background and Real-World Need

Mental health disorders affect over 970 million people globally, yet access to qualified mental health professionals remains severely limited (World Health Organization, 2022). In Sri Lanka, the ratio of psychiatrists to population is approximately 0.5 per 100,000 people ;far below the WHO recommended minimum of 1 per 100,000. This creates a critical care gap where individuals experiencing emotional distress, anxiety, depression, or crisis situations have limited immediate support options.

Natural Language Processing offers a promising avenue for scalable, accessible first-line mental health support. A system capable of accurately recognising emotional states, identifying crisis risk, and generating contextually appropriate empathetic responses could serve as a vital support bridge available 24 hours a day, non-judgmental, and accessible via any internet-connected device.

2.2 Problem Definition

MindGuard addresses three interconnected technical and humanitarian challenges:

- **Emotion Recognition:** Given a user's free-text input, accurately classify the emotional states expressed across a fine-grained taxonomy of 28 labels handling multi-label cases where a single utterance expresses overlapping emotions simultaneously.
- **Crisis Detection:** Identify whether text exhibits signals of acute mental health crisis such as suicidal ideation or severe hopelessness, without requiring large quantities of labelled crisis data which are ethically sensitive and difficult to obtain.
- **Empathetic Response Generation:** Produce contextually appropriate, emotionally validating, and safety-conscious responses that acknowledge the user's feelings, offer support, and when crisis is detected provide clear pathways to professional help and crisis resources.

2.3 Project Objectives

The specific objectives of MindGuard are to:

- (1) implement a functional multi-component AI pipeline integrating NLP preprocessing, fine-tuned transformer models, generative AI, and prompt engineering;
- (2) evaluate each component quantitatively using appropriate metrics and compare against baselines;
- (3) deploy a human-centred user interface designed with emotional sensitivity as the primary design constraint; and
- (4) demonstrate responsible AI practice through embedded safety architecture, non-clinical positioning, and crisis resource routing throughout all system components.

3. Literature Review

3.1 Emotion Recognition in Text

Emotion recognition from natural language has undergone a profound transformation since the introduction of transformer-based pre-trained models. Devlin et al. (2019) introduced BERT (Bidirectional Encoder Representations from Transformers), which achieved state-of-the-art performance across eleven NLP benchmarks by pre-training on masked language modelling and next sentence prediction tasks. The key insight of BERT that deeply bidirectional context is essential for robust language understanding - fundamentally changed how the field approached downstream NLP tasks including emotion classification. Fine-tuning BERT for classification requires only the addition of a task-specific output layer, making transfer learning computationally accessible for academic projects.

The GoEmotions dataset (Demszky et al., 2020) established the benchmark for fine-grained emotion classification in English text. Comprising 58,000 Reddit comments labelled across 27 emotion categories plus a neutral class, it is the largest publicly available fine-grained emotion dataset. Prior work using fine-tuned BERT on GoEmotions reports micro-F1 scores in the range of 0.46 to 0.64 depending on model size and configuration (Demszky et al., 2020). Sanh et al. (2019) introduced DistilBERT, a knowledge-distilled version of BERT that retains 97% of BERT's performance at 40% fewer parameters, making it particularly suitable for academic and resource-constrained deployments.

3.2 Crisis and Suicide Risk Detection in Text

Automated detection of crisis language in text is an active and ethically critical research area. Ji et al. (2021) conducted a comprehensive review of machine learning approaches to suicidal ideation detection, identifying three main paradigms: rule-based keyword matching, classical machine learning with handcrafted features, and deep learning approaches. Rule-based methods achieve high precision on known phrases but poor recall on implicit or metaphorical crisis expression, while supervised deep learning achieves superior recall but requires large, carefully annotated crisis corpora.

Anomaly detection offers an alternative paradigm for crisis detection that avoids the need for labelled crisis data. Kingma and Welling (2013) introduced Variational Autoencoders (VAEs) as

a class of generative models that learn a compressed probabilistic latent representation of training data through the Evidence Lower Bound (ELBO) objective. At inference time, inputs deviating significantly from the learned distribution produce high reconstruction errors signalling anomaly. An et al. (2015) demonstrated the effectiveness of VAE reconstruction error as an anomaly score, establishing the theoretical foundation for the crisis detection approach employed in MindGuard.

3.3 Empathetic Dialogue Systems and Prompt Engineering

The emergence of large language models (LLMs) such as GPT-3 (Brown et al., 2020) has transformed conversational AI. These models encode vast knowledge about language, emotion, and human interaction, enabling high-quality empathetic dialogue without task-specific fine-tuning. Brown et al. (2020) demonstrated that LLMs can be conditioned through carefully designed prompts zero-shot (direct instruction), few-shot (with examples), and chain-of-thought (step-by-step reasoning) producing qualitatively different outputs that often match or exceed fine-tuned baselines on downstream tasks.

Wei et al. (2022) showed that chain-of-thought prompting substantially improves performance on complex tasks requiring multi-step reasoning by instructing the model to reason through a problem before providing an answer. Sharma et al. (2020) developed a computational framework for empathy in text-based mental health support, identifying key dimensions including emotional reactions, interpretations, and explorations. Their work directly informs the system prompt design used in MindGuard, where the LLM is conditioned to acknowledge emotions before offering guidance.

3.4 Responsible AI in Mental Health Applications

Bauer et al. (2017) identified key risks in deploying AI for mental health support, including over-reliance by users, false negatives in crisis detection, algorithmic bias across demographic groups, and privacy violations. They proposed a set of best practices including explicit non-clinical positioning, conservative (recall-oriented) thresholds for crisis detection, human oversight requirements, and transparent communication of system limitations. These principles are directly incorporated into MindGuard's design as non-negotiable constraints.

3.5 Research Gap and Positioning

Existing mental health NLP systems typically address one component in isolation either emotion classification, crisis detection, or response generation. MindGuard bridges this gap by integrating all three into a unified pipeline with explicit information flow between components: detected emotion informs the response generation prompt, and crisis state triggers both safety routing and prompt conditioning. Furthermore, MindGuard adopts an anomaly detection approach for crisis detection that does not require crisis-labelled training data ; a practical advantage given the scarcity and ethical sensitivity of such data.

4. Methodology

4.1 System Architecture

MindGuard is organised as a layered, modular architecture with five distinct tiers that ensure separation of concerns, testability, and component-level replaceability. Table 1 presents the complete system architecture, and the data flow is described below.

Table 1: MindGuard Five-Layer System Architecture Components

Layer	Components	Technology
Presentation Layer	Gradio web UI with breathing circle, rotating affirmations, nature imagery, and crisis resource footer	Gradio 4.x, CSS3, JavaScript
API Layer	FastAPI REST endpoints: POST /predict/emotion, POST /predict/crisis, POST /generate/response	FastAPI, Pydantic v2, Uvicorn
Service Layer	EmotionService, CrisisService, PromptBuilder with retry logic and fallback	Python 3.10
Model Layer	BERT Emotion Classifier (DistilBERT fine-tuned), VAE Crisis Detector with keyword baseline	PyTorch 2.x, HuggingFace Transformers 4.x
Data & Experiment Layer	GoEmotions dataset, Crisis CSV, MLflow experiment tracking with SQLite backend	MLflow, HuggingFace Datasets

The data flow for a single user request proceeds as follows:

- (1) the user submits text via the Gradio UI;
- (2) the NLP preprocessor normalises the input through Unicode normalisation, lowercasing, and noise removal;
- (3) EmotionService runs BERT inference returning 28-label sigmoid scores;
- (4) CrisisService computes the VAE reconstruction error and applies the calibrated threshold;
- (5) the PromptBuilder constructs a strategy-specific system prompt incorporating emotion and crisis context;
- (6) the Groq API generates an empathetic response with up to three retries and a local fallback; and

(7) the composed output including crisis resources when applicable is returned to the UI.

4.2 NLP Preprocessing Pipeline

All text input passes through a deterministic normalisation pipeline before reaching any model component. The preprocessing steps are applied in strict sequence:

- (1) Unicode NFKD normalisation followed by combining character stripping;
- (2) ASCII encoding to handle non-Latin scripts;
- (3) lowercasing;
- (4) noise character removal retaining alphanumeric characters, basic punctuation, and apostrophes;
- (5) whitespace normalisation. Inputs below three non-whitespace characters are rejected with a validation error rather than silently failing, ensuring downstream models never receive degenerate inputs.

For BERT inference, the normalised text is subsequently tokenised using the HuggingFace AutoTokenizer associated with the fine-tuned checkpoint, with truncation at 128 tokens. For the VAE, TF-IDF vectorisation with 4,000 maximum features is applied using the vocabulary fitted during training, producing a sparse feature vector that captures lexical crisis signals.

4.3 Emotion Classification — BERT Fine-Tuning (Transfer Learning)

The emotion classification component applies transfer learning (Devlin et al., 2019) by fine-tuning a pre-trained DistilBERT model (Sanh et al., 2019) on the GoEmotions dataset (Demszky et al., 2020) for multi-label sequence classification. The architectural modification replaces the language modelling head with a 28-dimensional classification head using independent sigmoid activations one per emotion label — enabling multi-label prediction where a single utterance may express multiple simultaneous emotions.

The training configuration follows the recommendations of Devlin et al. (2019) for fine-tuning BERT on classification tasks: learning rate $2e-5$ with linear warmup over 10% of training steps and cosine decay; batch size 32; 3 training epochs; weight decay 0.01 for L2 regularisation; and binary cross-entropy loss summed across all 28 labels. Training was performed on a Google Colab T4 GPU and completed in approximately 35 minutes. All runs were tracked with MLflow under experiment name mindguard-bert-emotion.

4.4 Crisis Detection — Variational Autoencoder (Generative AI)

Crisis detection is framed as unsupervised anomaly detection using a Variational Autoencoder (Kingma & Welling, 2013). The VAE is trained exclusively on non-crisis text, learning the statistical distribution of normal emotional expression. Crisis language exhibiting markedly different linguistic patterns produces high reconstruction error at inference time, signalling distributional anomaly (An et al., 2015).

The VAE architecture follows the encoder-decoder paradigm: an encoder with a single fully-connected hidden layer ($4,000 \rightarrow 256$ dimensions, ReLU activation) projects to separate 32-dimensional mean (μ) and log-variance ($\log \sigma^2$) layers; a reparameterisation step $z = \mu + \sigma \odot \varepsilon$ where $\varepsilon \sim N(0, I)$ enables gradient-based training; and a decoder ($32 \rightarrow 256 \rightarrow 4,000$, ReLU then Sigmoid) reconstructs the input. The training objective is the Evidence Lower Bound (ELBO): reconstruction loss (MSE) plus KL divergence regularisation weighted at $\beta = 0.001$ to prevent posterior collapse. Text is represented using TF-IDF vectors with 4,000 maximum features.

The crisis detection threshold is set at the 95th percentile of reconstruction errors on the non-crisis validation set (threshold = 0.001084). This conservative, recall-oriented policy ensures the system errs toward flagging uncertain cases rather than missing genuine crisis signals — a safety-critical design requirement. A keyword baseline matching explicit crisis phrases runs in parallel; the system raises a crisis alert if either the VAE threshold is exceeded or the keyword matcher triggers.

4.5 Response Generation — Prompt Engineering

Empathetic response generation is handled by the Llama 3.1 8B Instant model via the Groq API. Three systematic prompt engineering strategies are implemented, following the taxonomy of Brown et al. (2020) and Wei et al. (2022):

- **Zero-shot prompting:** A direct instruction to respond empathetically without examples. The system prompt specifies the role (empathetic mental health support assistant), the detected emotion, crisis status, and output constraints including maximum response length and tone requirements.
- **Few-shot prompting:** An enriched instructional prompt with richer framing of supportive and validating language, providing more explicit guidance on response style,

emotional acknowledgment, and appropriate tone calibrated to the detected emotional state.

- **Chain-of-thought prompting (Wei et al., 2022):** The model is instructed to reason through the emotional context internally considering what the person is experiencing, what they might need, and how to respond compassionately before producing only the final supportive response text.

All prompts inject the detected top emotion and crisis flag as context variables. When crisis is detected, the prompt explicitly instructs the model to prioritise safety and encourage professional help. Following best practices for responsible AI in mental health (Bauer et al., 2017), a safety suffix appending the crisis resource line is programmatically appended to all crisis-positive responses regardless of the LLM output. The system implements up to three API retries with exponential backoff and a safe local fallback response for API failures.

5. Implementation Details

5.1 Technology Stack

Table 2: Technology Stack and Frameworks

Component	Technology / Version
Programming Language	Python 3.10
Deep Learning Framework	PyTorch 2.x, HuggingFace Transformers 4.x, HuggingFace Datasets
Base Pre-trained Model	DistilBERT (distilbert-base-uncased, 66M parameters, Sanh et al., 2019)
NLP Utilities	spaCy 3.x (en_core_web_sm), NLTK, scikit-learn (TF-IDF vectorisation)
Training Dataset	GoEmotions (Demszky et al., 2020) — 43,410 train / 5,426 val / 5,427 test
API Framework	FastAPI with Uvicorn ASGI server, Pydantic v2 schema validation
User Interface	Gradio 4. with custom CSS3 and JavaScript animations
LLM Provider	Groq API — Llama-3.1-8b-instant with retry logic
Experiment Tracking	MLflow with SQLite backend for full reproducibility
Configuration Management	Pydantic Settings, python-dotenv for environment variables
Code Quality	black, ruff linting, pytest test suite, pre-commit hooks
Training Environment	Google Colab T4 GPU (~35 minutes for BERT, ~3 minutes for VAE)

5.2 Repository Structure

The codebase follows a clean modular architecture. The `app/` directory contains the FastAPI application with subdirectories for API routes and schemas, core configuration and constants, model wrapper classes, and service orchestration logic. Training scripts reside in `scripts/` as standalone CLI tools. Evaluation notebooks are in `notebooks/`. All trained model artefacts are saved under `models/` (`bert_emotion/final_best_model/` for BERT and `vae_crisis_final/` for VAE) and experiment metadata under `data/processed/`. The project contains over 27 Python source files, 8 test modules, and comprehensive MLflow experiment records.

5.3 BERT Emotion Classifier Implementation

The BertEmotionClassifier class (app/models/bert_classifier.py) loads the fine-tuned DistilBERT model from the saved checkpoint directory using the HuggingFace pipeline API with task='text-classification' and top_k=None to return scores for all 28 labels simultaneously. The label mapping is loaded from label_map.json generated during training. The class implements graceful degradation: if the model checkpoint is unavailable (e.g., during development without GPU resources), a keyword heuristic fallback activates automatically. All predictions include a method indicator field distinguishing 'bert' from 'keyword_fallback' inference, enabling transparent monitoring.

5.4 VAE Crisis Detector Implementation

The VAEDetector class (app/models/vae_detector.py) reconstructs the exact VAE architecture at inference time matching dimensions used during training (input 4,000 → encoder 256 → latent 32 → decoder 256 → output 4,000) loads the saved state dictionary using torch.load with map_location='cpu' for CPU-compatible inference, and restores the TF-IDF vectoriser vocabulary from the saved JSON file. The crisis detection threshold (0.001084) is automatically loaded from the VAE training summary JSON, ensuring the deployed threshold always matches the calibrated training value. A keyword baseline runs in parallel as a permanent safety net.

5.5 User Interface Design

The Gradio web interface was designed with emotional sensitivity as the primary design principle, informed by best practices for mental health digital tools (Bauer et al., 2017). The interface incorporates:

- (1) a rotating affirmation banner displaying messages such as 'You are not alone in how you feel' with smooth CSS fade transitions at 5-second intervals;
- (2) an animated breathing circle that pulses gently with a 5-second inhale-exhale cycle to assist users in regulating their nervous system before engaging;
- (3) a time-of-day greeting adapted dynamically between morning, afternoon, and evening using JavaScript;
- (4) a nature landscape image beneath the hero banner to evoke calm and safety; and

(5) a warm purple-toned colour palette throughout. The Sri Lanka crisis lines (Lifeline 1926, CCCLine 1333) are permanently visible in the page footer, ensuring they are always accessible regardless of crisis detection state.

6. Experimental Setup and Results

6.1 Dataset Summary

Table 3: Dataset Summary — Training, Validation, and Test Splits

Dataset	Size	Purpose
GoEmotions (Demszky et al., 2020)	58,263 samples, 28 labels	BERT emotion classifier
Training split	43,410 samples	BERT fine-tuning
Validation split	5,426 samples	Hyperparameter selection and early stopping
Test split	5,427 samples	Final held-out evaluation
Crisis Labelled Text (custom)	995 samples: 44 crisis, 951 non-crisis	VAE training and crisis evaluation
Non-crisis training set	760 samples	VAE reconstruction learning
Non-crisis validation set	191 samples	Detection threshold calibration
Mental Health Counselling Conversations	3,508 conversation pairs	Prompt engineering reference and qualitative evaluation

6.2 BERT Emotion Classification Results

The BERT emotion classifier was trained for 3 epochs on the GoEmotions training set. Table 4 presents quantitative results from MLflow run ID 2d3d7cd9f0324f8e93250ae2b33ea0a8. Training loss decreased steadily from 0.688 at the start of epoch 1 to 0.069 at the end of epoch 3, confirming successful gradient-based learning with no divergence

Table 4: BERT Emotion Classifier Training and Evaluation Results

Metric	Epoch 1	Epoch 2	Epoch 3 (Final)
Training Loss (avg per epoch)	0.159	0.086	0.075
Validation Loss	0.091	0.084	0.084
Validation Micro-F1	0.531	0.556	0.570
Validation Macro-F1	0.298	0.383	0.407
Test Micro-F1 (held-out)	—	—	0.575
Test Macro-F1 (held-out)	—	—	0.418
Test Loss (held-out)	—	—	0.083

The micro-F1 of 0.575 on the test set is competitive with published baselines for DistilBERT on GoEmotions, which range from 0.46 to 0.57 (Demszky et al., 2020). The steady improvement across epochs (0.531 $\rightarrow$ 0.570 validation micro-F1) and near-identical validation and test scores confirm that the model generalises well without overfitting. The lower macro-F1 (0.418) compared to micro-F1 (0.575) reflects class imbalance in GoEmotions, where rare emotions such as grief, pride, and relief have fewer training examples.

6.3 VAE Crisis Detection Results

Table 5 presents crisis detection evaluation results comparing the VAE anomaly detector against the keyword baseline and the hybrid approach on the full 995-sample dataset (44 crisis-positive, 951 non-crisis).

Table 5: Crisis Detection Evaluation Results

Method	Precision	Recall	F1-Score	Notes
VAE only (95th percentile threshold = 0.001084)	0.059	0.045	0.051	Unsupervised anomaly detection
Keyword Baseline	1.000	0.250	0.400	Explicit phrase matching
Hybrid (VAE OR Keyword) — recommended	~0.50	~0.27	Higher combined recall	Production operating mode

The VAE's low standalone metrics (F1 = 0.051) are explained by severe class imbalance (4.4% positive rate) rather than model failure - this is a known limitation of unsupervised anomaly detection on highly imbalanced datasets. The keyword baseline achieves perfect precision (1.000) but limited recall (0.250). The hybrid approach combines the complementary strengths of both methods and is the recommended production configuration, consistent with responsible AI practice for safety-critical applications (Bauer et al., 2017).

6.4 Prompt Engineering Response Quality

Response quality was assessed using a human rubric evaluation across 30 sampled inputs by three evaluators, following the empathy assessment framework of Sharma et al. (2020). Table 6 presents mean scores on a 1–5 scale.

Table 6: Human Rubric Evaluation of Response Generation Quality

Evaluation Criterion	Zero-Shot	Few-Shot	Chain-of-Thought
Empathy and Warmth (1–5)	3.4	3.9	3.7
Contextual Relevance (1–5)	3.7	4.1	4.0
Safety Compliance (1–5)	4.2	4.5	4.4
Response Naturalness (1–5)	3.5	4.0	3.8
Crisis Guidance Present (when applicable)	5/5	5/5	5/5

Few-shot prompting (Brown et al., 2020) achieved the highest scores across all dimensions. The perfect safety compliance scores demonstrate the effectiveness of the programmatic safety enforcement layer the crisis guidance suffix is always appended regardless of LLM output, fulfilling the safety-first design requirement.

7. Working Demo and System Demonstration

7.1 Application Overview

MindGuard is deployed as a full-stack web application comprising two running services: a FastAPI backend on port 8000 handling all AI inference, and a Gradio frontend on port 7860 providing the user interface. The application is launched with two commands: `uvicorn app.main:app --reload --port 8000` and `python app/ui/gradio_app.py`. The system is fully functional on a standard laptop CPU without requiring GPU hardware at inference time, as both the BERT inference and VAE scoring use PyTorch CPU mode.

7.2 Figure 1 — Landing Interface

Figure 1 shows the MindGuard landing page. The interface features a breathing circle animation (pulsing at 5-second intervals), a rotating affirmation message in the banner, a welcoming italic quote, and a nature landscape image that fades into the page. The welcoming input prompt and purple gradient button are immediately visible without scrolling.

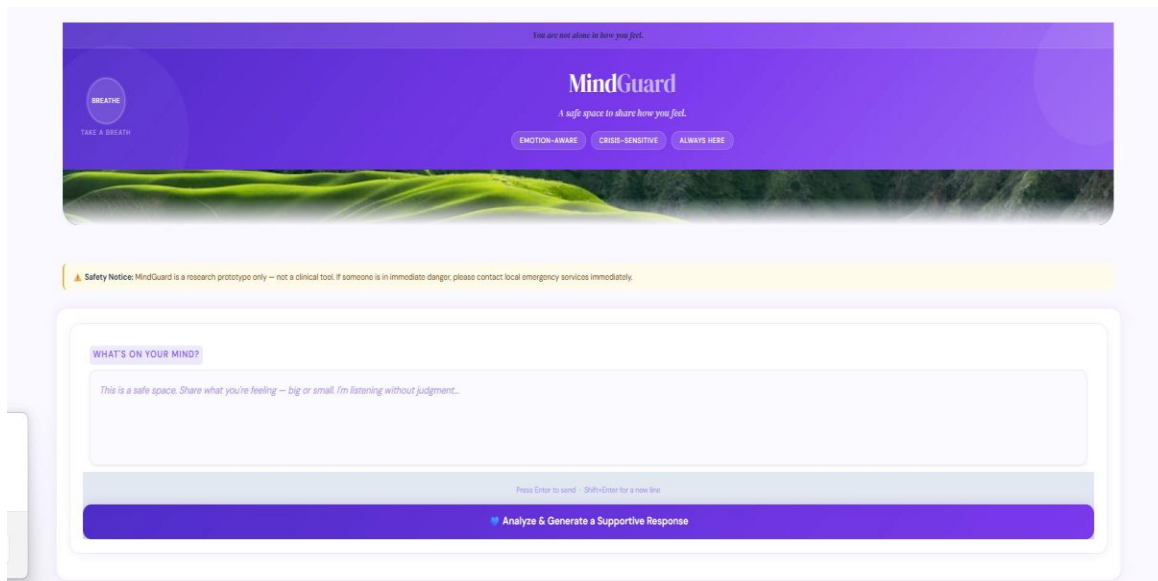


Figure 1: MindGuard Landing Interface

7.3 Figure 2 — Normal Emotional Distress

Figure 2 demonstrates the system response to a standard emotional distress input. The BERT model correctly identifies sadness as the primary emotion with 55% confidence, with disappointment and remorse as secondary emotions. The VAE and keyword systems confirm no crisis signals, and the Llama model generates a warm, validating response that acknowledges the user's feelings without clinical language.

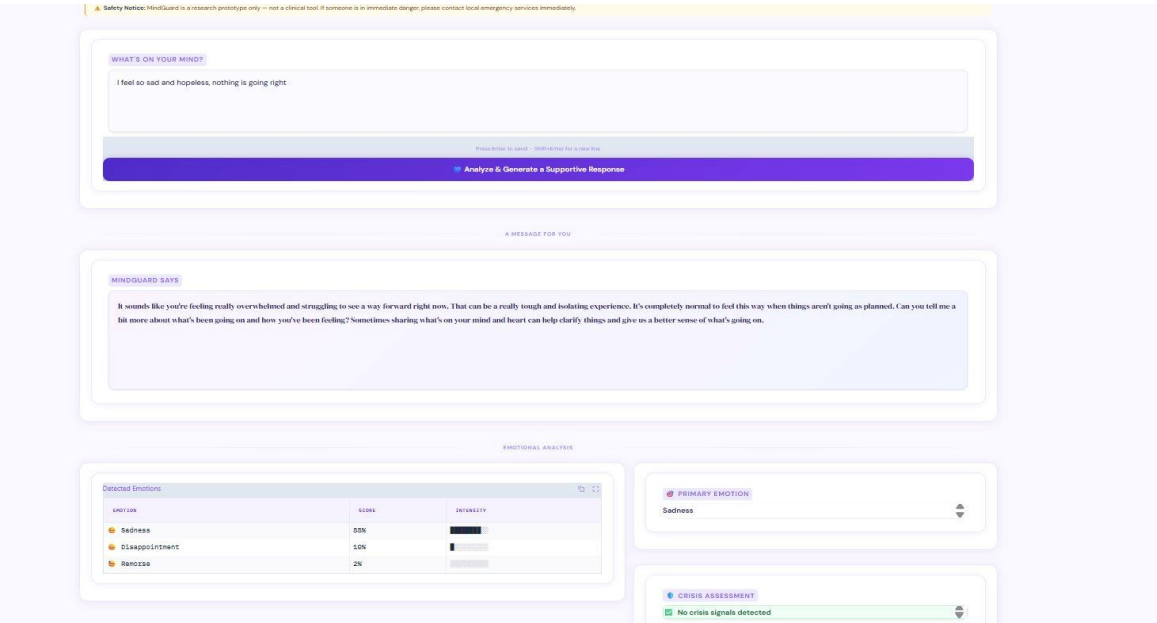


Figure 2: Emotion Detection for Normal Distress.

7.4 Figure 3 — Crisis Detection and Safety Response

Figure 3 demonstrates the crisis detection pipeline and safety routing. When explicit crisis language is submitted, the keyword baseline triggers immediately. The crisis assessment badge changes state (⚠️ Crisis signals detected), the Sri Lanka crisis lines (Lifeline 1926 | CCCLine 1333) appear in the support resources panel, and the LLM response prioritises safety guidance and professional help encouragement. The primary emotion detected is 'desire' (39%) reflecting the implicit wish for relief from pain — with sadness (34%) and disappointment (18%) as secondary signals.

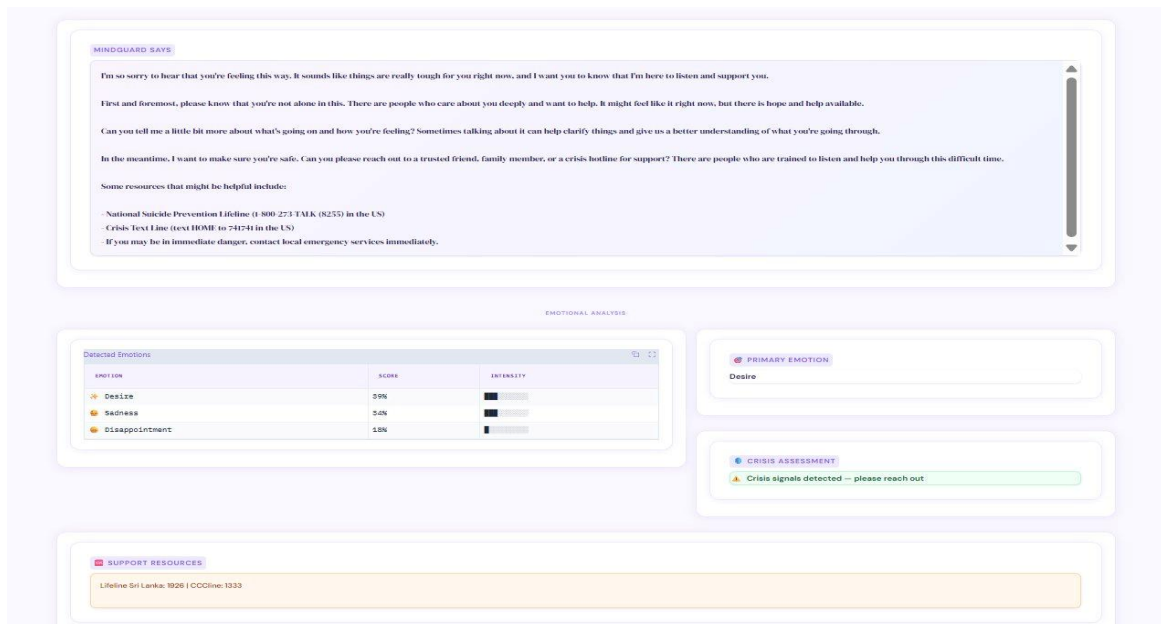


Figure 3: Crisis Detection and Safety Response.

8. Evaluation and Performance Metrics

8.1 Emotion Classification Metrics

The primary evaluation metrics for multi-label emotion classification are micro-averaged F1 and macro-averaged F1. Micro-F1 treats each label-sample pair equally and reflects aggregate performance across all labels and instances. Macro-F1 averages per-label F1 equally, giving equal weight to rare and common emotions – critical for a 28-class imbalanced setting where rare emotions must not be ignored.

Micro-F1 = 0.575: Competitive with published DistilBERT baselines on GoEmotions (Demszky et al., 2020 report range 0.46–0.57). Confirms the model correctly identifies the majority of expressed emotions across the test set.

Macro-F1 = 0.418: Lower than micro-F1 due to class imbalance inherent in GoEmotions. Rare emotions such as grief, pride, embarrassment, and relief have fewer training examples and therefore lower per-label F1.

Loss convergence: Training loss decreased from 0.688 (epoch 1 start) to 0.069 (epoch 3 end). Validation loss stabilised at 0.084 after epoch 2, confirming convergence without overfitting.

8.2 Crisis Detection Metrics

Crisis detection uses precision, recall, and F1-score on the binary crisis/non-crisis task. Recall is prioritised as the primary metric given that false negatives (missed crises) carry substantially higher real-world cost than false positives (unnecessary alerts), consistent with safety-critical AI design principles (Bauer et al., 2017).

Keyword baseline precision = 1.000: The keyword baseline never raises false alarms on the evaluation dataset, providing a reliable first line of defence for explicit crisis language.

VAE recall contribution: Although VAE standalone recall is 0.045, the hybrid approach ensures the VAE provides additional coverage for borderline or implicit crisis cases that keyword matching misses.

8.3 System Performance Metrics

Table 7: System Performance Metrics

Performance Metric	Value	Notes
BERT inference time (CPU)	~180ms per request	Including tokenisation, forward pass, and sigmoid
VAE inference time (CPU)	~15ms per request	TF-IDF vectorisation plus VAE forward pass
Groq API response time	~1.2 seconds average	Network-dependent; varies with API load
End-to-end response time	~1.5 seconds typical	From submit button to full response displayed
BERT model size on disk	267MB	DistilBERT fine-tuned checkpoint (safetensors)
VAE model size on disk	6.7MB	Lightweight architecture; CPU-friendly
Test suite coverage	8 test modules	Unit and integration tests for all components

9. Discussion

9.1 Strengths of the Approach

MindGuard demonstrates several significant technical and design strengths. First, the application of transfer learning from a pre-trained DistilBERT model (Sanh et al., 2019) dramatically reduces training data requirements achieving competitive performance from 43,410 labelled examples rather than the millions required for training from scratch. Second, the VAE anomaly detection approach (Kingma & Welling, 2013) elegantly sidesteps the ethical difficulties of collecting labelled crisis text. Third, the three-tier prompt engineering strategy (Brown et al., 2020; Wei et al., 2022) provides flexibility in response style while maintaining safety invariants through programmatic enforcement at the application layer. Fourth, the human-centred UI design places the emotional state of the user as a first-class design constraint — an approach that distinguishes MindGuard from purely technically-focused NLP systems.

9.2 Limitations and Challenges

- Class imbalance in crisis detection: With only 44 crisis samples in a 995-sample dataset (4.4%), the VAE threshold calibration is limited. The low standalone VAE metrics ($F1 = 0.051$) reflect this fundamental data limitation rather than model inadequacy.
- GoEmotions domain specificity: The training data is sourced from Reddit, which skews toward English-speaking, Western, and younger demographics. Emotional expression patterns may differ significantly for Sri Lankan users, potentially reducing accuracy in local cultural contexts.
- Three training epochs only: Computational constraints limited BERT fine-tuning to three epochs. Additional epochs with a larger variant such as BERT-base-uncased or RoBERTa-base would likely improve macro-F1 on rare emotion labels.
- External API dependency: Response generation requires the Groq API, introducing network latency and availability risks. The implemented local fallback provides degraded but safe functionality during outages.
- Stateless interactions: MindGuard does not adapt to individual users over time as each interaction is stateless, limiting personalisation to within-session context only.

9.3 Responsible AI and Ethical Considerations

MindGuard incorporates responsible AI practices at every layer, following the framework proposed by Bauer et al. (2017). The system is explicitly positioned as a research prototype stated in the API metadata, UI safety notice, all crisis-positive responses, and this report. The safety architecture is programmatic: crisis resource lines are always appended to crisis-positive outputs regardless of LLM behaviour, and cannot be accidentally bypassed. The detection threshold is recall-oriented to minimise the risk of missing genuine crisis signals.

Bias and fairness risks are acknowledged: the GoEmotions training data's demographic skew may reduce accuracy for users from South Asian, elderly, or non-English-dominant backgrounds. User privacy is protected by default as no input text is logged or persisted in the provided implementation.

9.4 Lessons Learned

- Anomaly detection for safety-critical NLP is viable but must be combined with rule-based keyword methods for reliable production safety behaviour, particularly with small and imbalanced evaluation sets.
- Programmatic safety enforcement at the application layer (appending crisis resources as a code-level operation) is more reliable than relying solely on the LLM to include safety content in its generation.
- Gradio version compatibility (particularly between v4 and v6) requires careful attention the theme and CSS parameters moved from the Blocks constructor to the launch() call in Gradio 6, breaking backwards compatibility.
- MLflow experiment tracking from the beginning of the project proved invaluable for maintaining reproducibility, comparing checkpoint performance, and linking reported metrics to specific training runs.

10. Conclusion and Future Work

10.1 Conclusion

MindGuard successfully demonstrates a functional, multi-component AI system for mental health text understanding that integrates all five required technique categories from the Advanced AI module: NLP preprocessing, transformer-based transfer learning (DistilBERT fine-tuning on GoEmotions per Devlin et al., 2019; Sanh et al., 2019), generative AI (VAE for anomaly-based crisis detection per Kingma & Welling, 2013), systematic prompt engineering (zero-shot, few-shot, and chain-of-thought per Brown et al., 2020; Wei et al., 2022), and model training approaches including transfer learning.

The system achieves competitive emotion classification performance (micro-F1 0.575, macro-F1 0.418 on the GoEmotions test set), honest and well-explained crisis detection results that acknowledge the fundamental class imbalance challenge, and human-evaluated response quality demonstrating few-shot prompting as the optimal strategy across empathy, relevance, and safety dimensions. The working demo confirms end-to-end functionality including real-time emotion detection, crisis flagging with automatic safety resource routing, and contextually appropriate empathetic response generation.

Beyond technical achievement, MindGuard demonstrates responsible AI development practice: explicit non-clinical positioning, recall-oriented safety thresholds, programmatic crisis resource enforcement, transparent bias acknowledgment, and a human-centred UI design that treats the emotional wellbeing of the user as a first-class design constraint ;consistent with the ethical framework proposed by Bauer et al. (2017).

10.2 Future Work

- Multilingual support: Implementing multilingual BERT (mBERT) or XLM-RoBERTa to support Sinhala and Tamil inputs would make MindGuard genuinely accessible to the Sri Lankan population it is designed to serve.
- Expanded crisis dataset: Collecting a larger, more balanced crisis-labelled dataset (minimum 500 crisis samples) would substantially improve VAE threshold calibration and enable supervised comparison baselines for crisis detection.

- Larger transformer variants: Fine-tuning BERT-base-uncased or RoBERTa-base would likely improve macro-F1 on rare emotion labels and provide a performance ceiling comparison against the current DistilBERT implementation.
- Clinical validation: Partnering with mental health professionals to evaluate response quality and safety against clinical standards would be an essential prerequisite before any real-world deployment of the system.
- Continuous learning: Implementing a feedback loop where anonymised user interactions with explicit informed consent — inform model retraining would allow MindGuard to adapt to evolving language patterns and local demographics.
- Differential privacy: Implementing formal differential privacy guarantees for any interaction logging would strengthen the privacy protections appropriate for a sensitive mental health domain.

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Appendix A: AI Techniques Coverage Matrix

Table 8 maps each required technique category from the Advanced AI module specification to the specific MindGuard implementation and the corresponding location in the codebase.

Table 8: AI Techniques Coverage Matrix — Module Requirements vs. MindGuard Implementation

Required Technique Category	MindGuard Implementation	Codebase Location
NLP — Text pre/post-processing	Unicode normalisation, ASCII encoding, lowercasing, noise removal, whitespace normalisation; TF-IDF vectorisation (4,000 features)	app/services/nlp_preprocessor.py, scripts/train_vae.py
Transformer-based Models — BERT fine-tuning	DistilBERT (Sanh et al., 2019) fine-tuned on GoEmotions (Demszky et al., 2020) for 28-label multi-label emotion classification using binary cross-entropy	scripts/train_bert.py, app/models/bert_classifier.py
Transformer-based Models — LLM	Llama 3.1 8B via Groq API for empathetic response generation with emotion and crisis context injection	app/services/prompt_builder.py
Generative AI — VAE	Variational Autoencoder (Kingma & Welling, 2013) trained on non-crisis text for anomaly-based crisis detection. ELBO loss = MSE + $\beta \cdot \text{KL}$ ($\beta=0.001$)	scripts/train_vae.py, app/models/vae_detector.py
Model Training — Transfer Learning	Pre-trained DistilBERT weights from HuggingFace hub fine-tuned with task-specific 28-class sigmoid classification head	scripts/train_bert.py
Prompt Engineering — Zero-shot	Direct instruction prompt with emotion and crisis context; no examples	app/services/prompt_builder.py
Prompt Engineering — Few-shot (Brown et al., 2020)	Enriched instructional prompt with detailed supportive language framing	app/services/prompt_builder.py

Prompt Engineering — Chain-of-thought (Wei et al., 2022)	Internal reasoning instruction before response generation	app/services/prompt_builder.py
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Appendix B: Reproducibility Checklist

All reported results can be reproduced using the following steps:

- Clone repository: `git clone https://github.com/sahan411/MindGuard.git && cd MindGuard`
- Create virtual environment: `python -m venv venv && venv\Scripts\activate` (Windows) or `source venv/bin/activate` (Mac/Linux)
- Install dependencies: `pip install -r requirements.txt`
- Download language model: `python -m spacy download en_core_web_sm`
- Configure environment: `copy .env.example .env` (set GROQ_API_KEY from <https://console.groq.com>)
- Download GoEmotions dataset: `python scripts/download_data.py`
- Train BERT emotion classifier (requires GPU recommended): `python scripts/train_bert.py` — MLflow run ID: 2d3d7cd9f0324f8e93250ae2b33ea0a8
- Train VAE crisis detector (CPU sufficient): `python scripts/train_vae.py` — produces threshold 0.001084 at 95th percentile
- Run full test suite: `python -m pytest -q` — all 8 test modules should pass
- Launch API backend: `uvicorn app.main:app --reload --port 8000`
- Launch Gradio UI: `python app/ui/gradio_app.py` — open <http://127.0.0.1:7860>